SGDAgent Reasoning Loop

Question: Where does the girl take the vampire after the exam?
(A) A playground. (B) A home.
(C) A principal room. (D) A park.

Agent (Step 1):
Calling tool **Global-Level Retrieval** with query "girl take vampire after exam".

Global-Level Retrieval:
Matched Events:
evt_0071 [47:10–50:30] score=6.2420, confidence=0.8605
bench, vampire, girl; actions: sitting_on, next_to; support: 3 segment(s)
matched participants: **ent_000032 = vampire** (confidence=0.98)
ent_000049 = girl (confidence=0.95)
key relation: **(girl, sitting_on, bench)** (confidence=0.97)
support segments: seg_0116, seg_0120, seg_0121
.....

Agent (Step 2):
Calling tool **Segment-Level Retrieval** with query "bench vampire girl under tree".

Segment-Level Retrieval :
Matched Segments:
seg_0116 [47:55–48:25] score=10.3411
entities: tree, person, bench; predicates: sitting_on, under
relations: (person, sitting_on, bench), **(bench, under, tree)**

seg_0120 [49:35–50:05] score=9.4169
entities: vampire, girl, bench; predicates: sitting_on
relation: **(girl, sitting_on, bench)**
.....

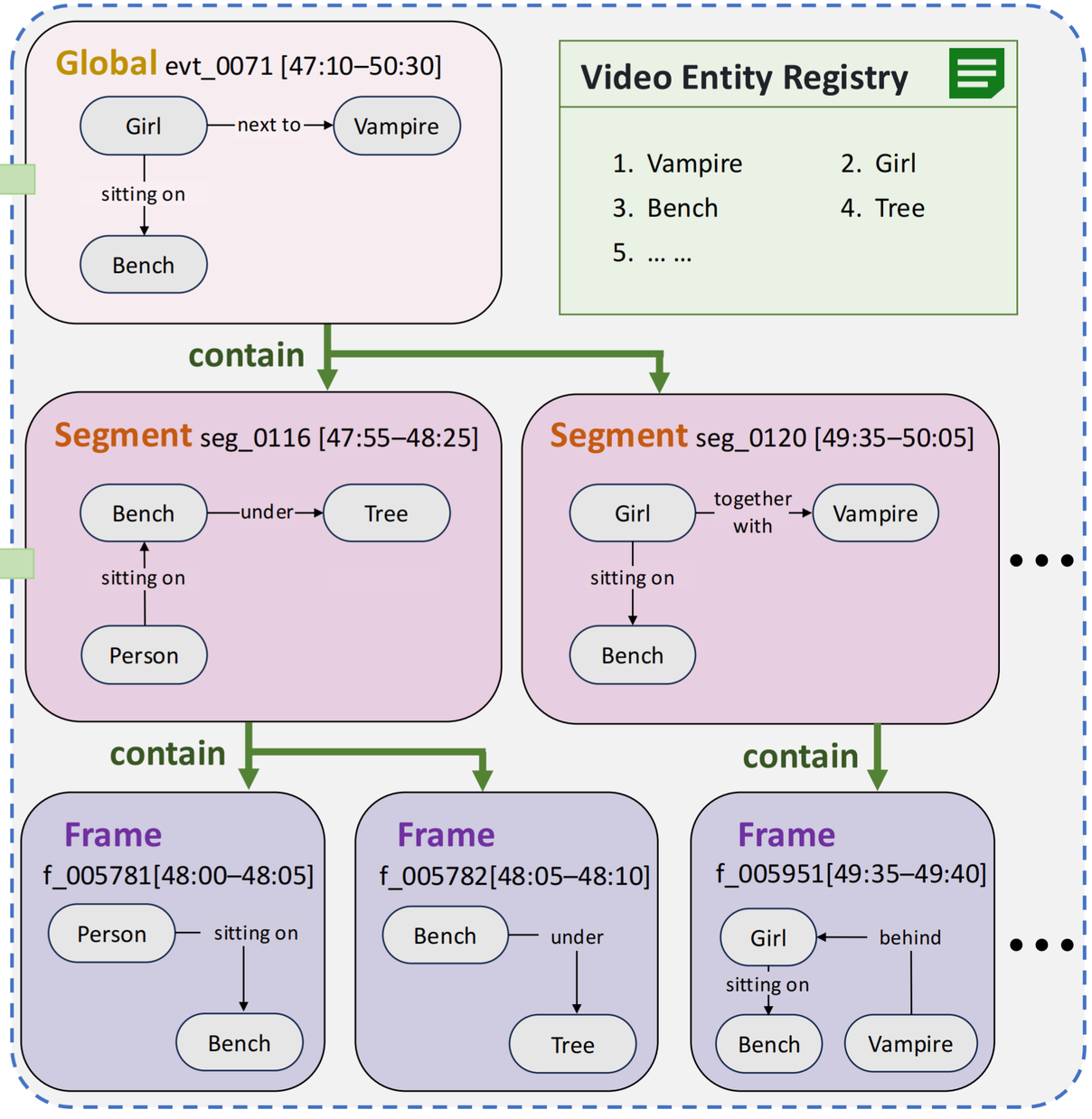
Agent (Step 3):
Calling tool **Frame Inspect** at ["00:48:10", "00:48:25"]:
"Describe the scene around the bench where the girl and vampire sit. Are there playground equipment, open lawn, or other indicators of park or playground?"

Frame Inspect:
Fine-grained Response:
The bench is outdoors, flanked by trees, with shrubs and distant greenery.
There is **no playground equipment.**
The scene looks like a **park or garden area**.

Agent (Step 4):
Executing **Answer**: (D) A park.



Long Video Input



Hierarchical Spatio-Temporal Scene Graph Representation